



## SCL Q3 - SCL QUESTION CALCULUS

CALCULUS - foundation (Universiti Teknologi Malaysia)



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$$y = \frac{1}{5} x^3, [0, 2] \quad \text{on } x\text{-axis}$$

If the curve  $y = f(x)$ ,  $a \leq x \leq b$  is rotated about the  $x$ -axis, then the surface area is given by

$$A = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$$

$$f(x) = \frac{1}{5} x^3 \quad f'(x) = \frac{3}{5} x^2$$

$$\begin{aligned} A &= 2\pi \int_0^2 \left(\frac{1}{5} x^3\right) \sqrt{1 + \left(\frac{3}{5} x^2\right)^2} dx \\ &= \frac{2\pi}{5} \int_0^2 x^3 \sqrt{1 + \frac{9}{25} x^4} dx \end{aligned}$$

$$\begin{aligned} \text{Let } u &= 1 + \frac{9}{25} x^4 \\ \frac{du}{dx} &= \frac{36}{25} x^3 \\ dx &= \frac{25}{36 x^3} du \end{aligned}$$

$$\begin{aligned} &= \frac{2\pi}{5} \int_0^2 \frac{25}{36} \sqrt{u} du \\ &= \frac{5\pi}{18} \left[ \frac{2}{3} u^{3/2} \right]_0^2 \\ &= \frac{5\pi}{18} \left[ \frac{2}{3} \left( 1 + \frac{9}{25} x^4 \right)^{3/2} \right]_0^2 \\ &= \frac{5\pi}{18} \left[ \left( \frac{2}{3} \left( 1 + \frac{9}{25} (2)^4 \right)^{3/2} \right) - \left( \frac{2}{3} \left( 1 + \frac{9}{25} (0)^4 \right)^{3/2} \right) \right] \\ &= \frac{5\pi}{18} \left( \frac{4394}{375} - \frac{2}{3} \right) \\ &= \frac{5\pi}{18} \left( \frac{4144}{375} \right) \\ &= \frac{2072\pi}{675} \quad \# \end{aligned}$$